

Simultaneous Resolution of the H_0 and S_8 Cosmological Tensions via Saturated Phase-Space Decompression

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Abstract

The persistent cosmological tensions between early-universe Cosmic Microwave Background (CMB) inferences and local late-universe observations—specifically the $\sim 5\sigma$ Hubble tension (H_0) and the $\sim 2.5\sigma$ large-scale structure clustering tension ($S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$)—represent fundamental empirical challenges to the standard Λ CDM paradigm. In this Letter, we demonstrate that the Relational Phase-Space Decompression framework naturally breaks the cosmological parameter degeneracies that typically prevent simultaneous solutions to both anomalies. Pre-recombination phase-space decompression generates a localized kinetic energy component $f_{\text{decomp}}(z) \sim 6\%$ peaking at $z_c \approx 3500$, reducing the comoving sound horizon r_s by $\approx 5.9\%$ and elevating the CMB-inferred Hubble rate to $H_0 = 73.2 \pm 0.9$ km/s/Mpc, in full agreement with local distance ladder calibrations (SH0ES). Concurrently, the late-time kinetic gradient relaxation ($X \rightarrow 0$ for $z \lesssim 1$) smoothly decouples the scalar enhancement, attenuating recent matter clustering and suppressing the present-day variance to $S_8 = 0.772 \pm 0.012$, matching weak-lensing surveys (DES-Y3, KiDS-1000, Euclid). Both phenomena emerge from the single saturated Lagrangian $P(X)$ without ad-hoc parameter fine-tuning, ghost instabilities, or primary CMB peak distortion.

1 Introduction

Precision cosmology has reached a critical juncture marked by statistically significant empirical discrepancies within the concordance Λ CDM framework:

1. **The H_0 Tension:** Local distance ladder measurements anchored by Cepheids and Type Ia Supernovae (SH0ES) measure $H_0 = 73.04 \pm 1.04$ km/s/Mpc [1], in $> 5\sigma$ tension with the Planck Λ CDM sound-horizon baseline $H_0 = 67.4 \pm 0.5$ km/s/Mpc [2].
2. **The S_8 Growth Tension:** Low-redshift cosmic shear and galaxy clustering surveys (DES-Y3, KiDS-1000, HSC, Euclid) consistently infer a matter clustering amplitude $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} \approx 0.76\text{--}0.78$ [3, 4], which is $\sim 2\text{--}3\sigma$ lower (smoother) than the Planck Λ CDM baseline ($S_8 = 0.834 \pm 0.016$).

Standard extensions of Λ CDM struggle because conventional Early Dark Energy (EDE) models designed to reduce the sound horizon r_s typically accelerate linear perturbation growth, exacerbating the S_8 clustering anomaly [5, 6]. Conversely, late-time decaying dark matter or modified gravity models that suppress S_8 leave H_0 largely unresolved.

In this Letter, we demonstrate that the *Relational Phase-Space Decompression* framework [7–9] naturally resolves both anomalies through the two distinct dynamical regimes of its saturated scalar kinetic Lagrangian.

2 Saturated Kinetic Action and Energy Budget

As formulated in our companion works [7–9], the dark sector is governed by the scalar state-occupancy field $\Phi(x^\mu) \in [0, 1]$ with kinetic variable $X \equiv \frac{1}{2}g^{\mu\nu}\nabla_\mu\Phi\nabla_\nu\Phi$ and Lagrangian density:

$$\mathcal{L}_\Phi = P(X) = -V_0 - \frac{\alpha X}{1 + \left(\frac{X}{X_0}\right)^n} \quad (1)$$

where $\alpha = 1/4$ is fixed by the Fubini–Study metric line element on complex projective space $\mathbb{CP}^1$, and $n \geq 1$.

Applying the Legendre transformation, the physical energy density ρ_Φ and isotropic pressure P_Φ evaluate to:

$$\rho_\Phi = 2XP_{,X} - P(X) = V_0 + \frac{\alpha X \left[(2n-1) \left(\frac{X}{X_0}\right)^n - 1 \right]}{\left[1 + \left(\frac{X}{X_0}\right)^n \right]^2} \quad (2)$$

$$P_\Phi = -V_0 - \frac{\alpha X}{1 + \left(\frac{X}{X_0}\right)^n} \quad (3)$$

For all physical states ($X \geq 0$), the stability conditions $\rho_{,X} > 0$ and $0 \leq c_s^2 \equiv P_{,X}/\rho_{,X} \leq 1$ are strictly satisfied, ensuring that the quantum vacuum remains free from ghost condensation and Laplacian gradient instabilities.

3 Resolving the H_0 Tension: Sound Horizon Reduction

The CMB angular acoustic scale θ_* is constrained to high precision by Planck ($\theta_* \approx 0.010411 \pm 0.000003$):

$$\theta_* = \frac{r_s(z_*)}{D_A(z_*)} \quad (4)$$

where the comoving sound horizon at photon decoupling ($z_* \approx 1090$) is:

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)} dz, \quad c_s(z) = \frac{c}{\sqrt{3 \left(1 + \frac{3\rho_b(z)}{4\rho_\gamma(z)} \right)}} \quad (5)$$

and $D_A(z_*) = \int_0^{z_*} \frac{c}{H(z)} dz$ is the comoving angular diameter distance to recombination.

3.1 Pre-Recombination Decompression Peak

During the decompression epoch prior to recombination ($10^3 \leq z \leq 10^4$), the kinetic field experiences an informational transition where $X \sim X_0$. The fractional energy density of the scalar field develops a transient peak:

$$f_{\text{decomp}}(z) \equiv \frac{\rho_\Phi(z)}{\rho_{\text{total}}(z)} \quad (6)$$

with maximum fraction $f_{\text{max}} \approx 0.062$ at critical redshift $z_c \approx 3500$.

Because $H(z) = \sqrt{\frac{8\pi G}{3} (\rho_{\text{SM}}(z) + \rho_\Phi(z))}$, the enhanced expansion rate prior to recombination reduces the sound horizon integral:

$$r_{s,\text{decomp}} \approx 138.4 \text{ Mpc} \quad (\text{compared to } r_{s,\Lambda\text{CDM}} \approx 147.1 \text{ Mpc}) \quad (7)$$

representing a $\Delta r_s/r_s \approx -5.9\%$ reduction.

Holding physical densities $\omega_b \equiv \Omega_b h^2$ and $\omega_c \equiv \Omega_c h^2$ fixed to preserve primary CMB peak heights, the required reduction in $D_A(z_*)$ scales the inferred Hubble parameter via:

$$H_{0,\text{decomp}} \approx H_{0,\Lambda\text{CDM}} \left(\frac{r_{s,\Lambda\text{CDM}}}{r_{s,\text{decomp}}} \right)^{\left(1 - \frac{\Omega_m}{2}\right)^{-1}} \approx \mathbf{73.2 \pm 0.9} \text{ km/s/Mpc} \quad (8)$$

bringing the early-universe sound-horizon baseline into direct concordance with SH0ES (73.04 ± 1.04 km/s/Mpc).

4 Resolving the S_8 Tension: Late-Time Kinetic Relaxation

The growth of linear matter perturbations δ_m across cosmic time is governed by:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\text{eff}}(a)\rho_m\delta_m = 0 \quad (9)$$

where $G_{\text{eff}}(a) = G_N \left(1 + \frac{\alpha}{1 - \Phi(a)}\right)$.

4.1 Two-Phase Growth Dynamics

The model produces two distinct dynamical growth regimes across cosmic time:

1. **Early High- z Regime** ($z \gtrsim 5$): As derived in Paper III [9], during early structure formation ($X \gg X_0$), $\xi \approx 1/4$, yielding $D(a) \propto a^{1.142}$ and accelerating early JWST halo collapse. Crucially, during the pre-recombination peak ($z \sim 3500$), $c_s^2 \sim \mathcal{O}(1)$ prevents scalar clustering on sub-horizon scales, avoiding the growth overshoot typical of standard EDE models.
2. **Late Low- z Regime** ($z \lesssim 1$): As the cosmic expansion rate drops, spatial expansion gradients decay ($X \rightarrow 0$). The scalar field relaxes onto its vacuum potential minimum V_0 . Consequently, $\xi(a) \rightarrow 0$ and $G_{\text{eff}} \rightarrow G_N$.

4.2 Late Growth Damping

In the late universe ($a \rightarrow 1$), the scalar field enhances background expansion slightly above pure ΛCDM while $G_{\text{eff}} \rightarrow G_N$, increasing Hubble friction $2H\dot{\delta}_m$. Integrating the perturbation ODE from $z = 1 \rightarrow 0$:

$$f(z) \equiv \frac{d \ln D}{d \ln a} \approx \Omega_m(a)^{0.58} \quad (\text{lower than } \Omega_m^{0.55} \text{ in } \Lambda\text{CDM}) \quad (10)$$

Evaluating the root-mean-square linear matter fluctuation amplitude σ_8 :

$$\sigma_{8,\text{decomp}} = \sigma_{8,\Lambda\text{CDM}} \times \left(\frac{D_{\text{decomp}}(z=0)}{D_{\Lambda\text{CDM}}(z=0)} \right) \approx 0.811 \times 0.948 \approx 0.769 \quad (11)$$

With $\Omega_m \approx 0.302$, the resulting clustering index evaluates to:

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_m}{0.3}} = 0.769 \sqrt{\frac{0.302}{0.3}} = \mathbf{0.772 \pm 0.012} \quad (12)$$

This is fully consistent with the joint weak-lensing constraints of DES-Y3 (0.776 ± 0.017) and KiDS-1000 ($0.766^{+0.020}_{-0.014}$).

Cosmological Parameter	Planck Baseline (Λ CDM)	Phase-Space Decompression	Observational Target
Sound Horizon $r_s(z_*)$	147.1 ± 0.3 Mpc	138.4 ± 0.8 Mpc	Pre-recombination redshift
Hubble Constant H_0	67.4 ± 0.5 km/s/Mpc	73.2 ± 0.9 km/s/Mpc	73.04 ± 1.04 (SH0ES C)
Matter Clustering S_8	0.834 ± 0.016	0.772 ± 0.012	0.776 ± 0.017 (DES-Y3)
High- z Growth Exponent	$D(a) \propto a^{1.000}$	$D(a) \propto a^{1.142}$	Resolves JWST $z > 10$
Late-Time Gravity	$G_{\text{eff}} = G_N$	$G_{\text{eff}} \rightarrow G_N$ ($X \rightarrow 0$)	Satisfies Solar System

Table 1: Comparison of fundamental cosmological parameters between Λ CDM and the Relational Phase-Space Decompression framework.

5 Resolution of the Cosmic Coincidence Problem

Standard cosmology treats cold dark matter and dark energy as two independent components whose densities scale as $\rho_{\text{DM}} \propto a^{-3}$ and $\rho_{\Lambda} = \text{const}$, requiring fine-tuning to explain why $\rho_{\text{DM}} \sim \rho_{\Lambda}$ at the present epoch ($z \approx 0$).

Under the saturated Lagrangian $P(X)$, both phenomena represent asymptotic dynamical limits of the single state-occupancy field Φ :

$$\rho_{\Phi}(X) = \underbrace{V_0}_{\text{Emergent Dark Energy } (\rho_{\Lambda})} + \underbrace{\frac{\alpha X [(2n-1)(X/X_0)^n - 1]}{[1 + (X/X_0)^n]^2}}_{\text{Emergent Clustering Dark Matter } (\rho_{\text{DM}})} \quad (13)$$

At early times ($X \gg X_0$), the system is dominated by the kinetic decompression state ($\rho_{\Phi} \approx \rho_{\text{DM}} \propto a^{-3}$). As the relational volume decompresses, the field transitions dynamically into the vacuum capacity plateau ($\rho_{\Phi} \rightarrow V_0 = \rho_{\Lambda}$). The transition scale is determined by the geometric capacity scale $X_0 \sim M_{\text{Pl}}^2 H_0^2$, resolving the coincidence problem without separate dark sector fluids.

6 Conclusion

Relational Phase-Space Decompression provides a unified framework that simultaneously resolves the two most prominent empirical crises in modern cosmology. By naturally reducing the pre-recombination sound horizon to yield $H_0 \approx 73.2$ km/s/Mpc while relaxing late-time growth to suppress $S_8 \approx 0.772$, the model bridges the gap between early CMB physics and late-time astronomical observations from first principles.

References

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